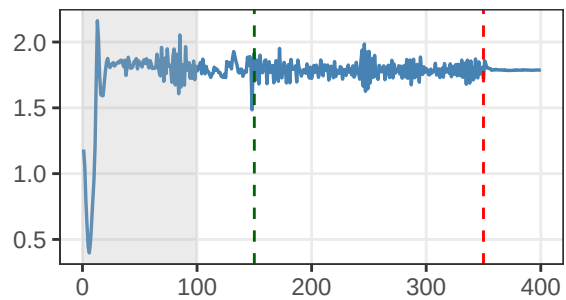
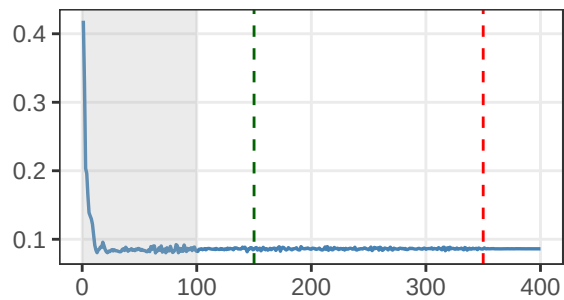
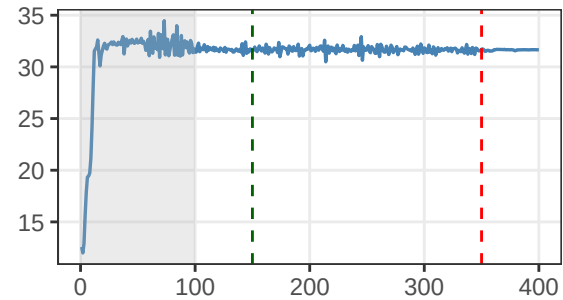
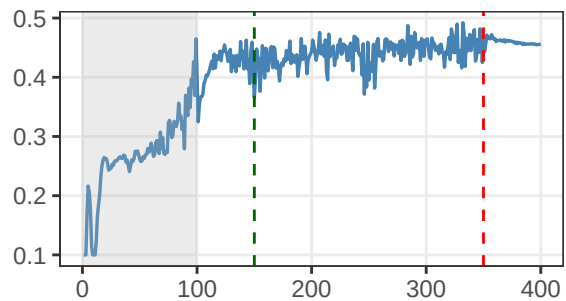
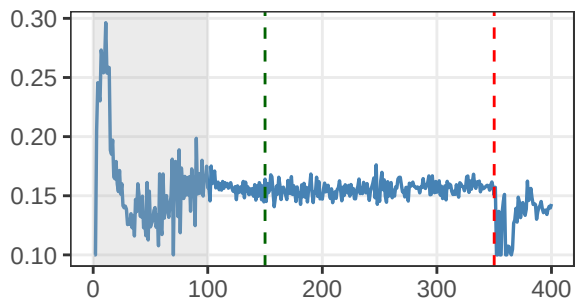
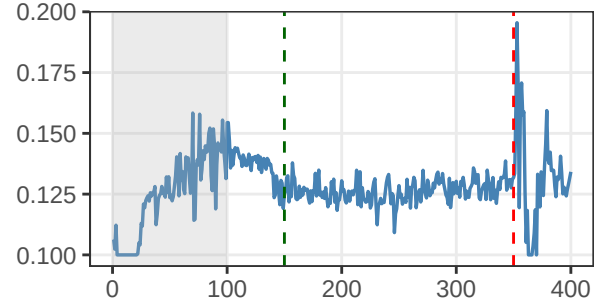
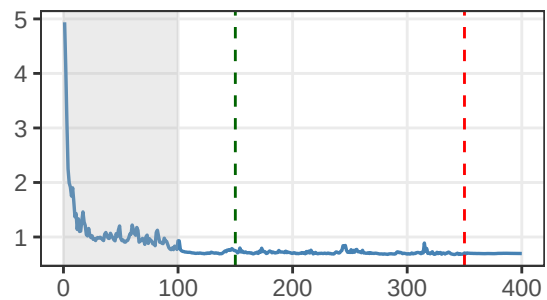


$k_{a,\text{pop}}$  $k_{e,\text{pop}}$  V_{pop}  ω_{k_a}  ω_{k_e}  ω_V  a  $-L_\psi(x)$ 